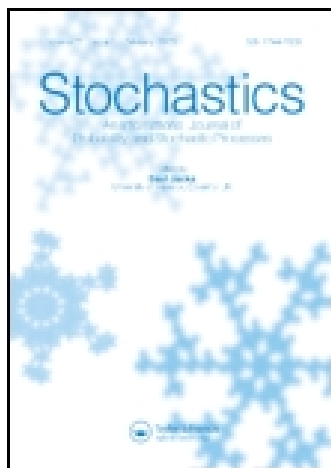




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The free energy of the dirac polaron, an explicit solution

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THE FREE ENERGY OF THE DIRAC POLARON, AN EXPLICIT SOLUTION

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A one-dimensional analog of the "Fröhlich-polaron" is considered. Using large deviation techniques we show that the asymptotics for the strong coupling theory of the polaron is given by a "mean field-polaron" and can be described by a variational problem whose solution is given explicitly.

KEY WORDS: Brownian motion, large deviation, polaron, variational problem.

AMS SUBJECT CLASSIFICATION: 60J65, 60F10, 49A34.

1. INTRODUCTION

The polaron problem consists of studying an electron which moves in an ionic crystal and polarizes the lattice in its neighborhood. The interaction changes the energy of the electron. Furthermore, when the electron moves, the polarization state must move with it. An electron moving with its accompanying distortion of the lattice is called a "polaron".

A special feature of the model is that the potential the electron feels at any given "time" depends on its position at previous "times" with a weighting factor. That is, the effect of the electron on the crystal propagates at a finite velocity and can make itself felt on the electron at a later time. This is less so for tighter binding, when the reaction of the crystal occurs much faster, what is referred to as "strong coupling".

Gross [9] presented a polaron model having the following Hamiltonian:

$$H = \frac{p^2}{2} + \alpha \sum_k (a_k^+ e^{-ikx} + a_k e^{ikx}) + \sum_k a_k^+ a_k$$

where x is the vector position of the electron, p its conjugate momentum, a_k^+ and a_k are the creation and annihilation operators of a phonon of momentum k . The constant α represents the coupling strength between the electron and the phonon field which depends on the dielectric constants of the ionic crystal.

Following the calculations of Feynman [8, (8.3)] and using the formal equality $\int_{\mathbb{R}} \exp\{ixy\} dy = \delta(x)$ we see that the lowest eigenvalue $E(\alpha)$ of the Gross Hamiltonian is given by

$$E(\alpha) = \lim_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \left\{ \alpha \int_0^t \int_0^t e^{-|\sigma-s|} \delta(X_\sigma - X_s) d\sigma ds \right\}.$$

We define

$$N(\lambda, t) = E^P \exp \left\{ \int_0^t \int_0^t \lambda e^{-\lambda|s-q|} \delta(X_s - X_\sigma) ds d\sigma \right\},$$

and

$$N(t) = E^P \exp \left\{ \frac{2}{t} \int_0^t \int_0^t \delta(X_s - X_\sigma) ds d\sigma \right\}$$

for all $\lambda > 0$ and $t > 0$, where P is the Wiener process on $\mathbb{R}$ starting at 0 and where δ is Dirac's function. We will show:

$$\lim_{\lambda \rightarrow \infty} \lim_{t \rightarrow \infty} \frac{1}{t} \log N(\lambda, t) = M, \quad (1)$$

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log N(t) = M, \quad (2)$$

where

$$M = \sup_{\substack{\phi \in L^2(\mathbb{R}) \\ \|\phi\|_2 = 1}} [2|\phi|_4^4 - \frac{1}{2}|\phi'|_2^2].$$

The variational problem can be solved explicitly and every solution is given by a translation of the even function

$$\phi(x) = \frac{\sqrt{2}}{\exp(-2x) + \exp(2x)}. \quad (3)$$

We call this one-dimensional model the Dirac polaron in order to stress the analogy with the Frölich polaron as treated in Feynman [8], Donsker and Varadhan [5] and Spohn [14]. Using (1) we can prove the Pekar conjecture for the Dirac polaron:

$$\lim_{\alpha \rightarrow \infty} \alpha^{-2} \lim_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \left\{ \int_0^t \int_0^t \alpha e^{-|s-\sigma|} \delta(X_s - X_\sigma) ds d\sigma \right\} = M. \quad (4)$$

(4) is a consequence of (1) after having applied the time rescaling $t' = t\sqrt{\alpha}$ to (4) and having used the following scaling properties of Brownian motion (BM) and Dirac's δ -function:

$$X(cs) = \sqrt{c} X(s) \text{ } P\text{-a.s.}; \quad \delta(c \cdot) = \frac{1}{c} \delta(\cdot).$$

This rescaling was introduced by Donsker and Varadhan and can be found at the beginning of Donsker and Varadhan [5, Section 4].

This article is organised as follows: In Section 2 we define the Dirac polaron in terms of a local time functional and prove a first approximation lemma. In Section 3 we treat the special feature of the Dirac polaron. This consists in an approximation procedure uniformly in λ which is different to what Donsker and Varadhan did in [5]. Donsker and Varadhan took advantage of a monotone approximation of the Coulomb potential by smooth functions. We have to use results about Brownian local time in order to obtain the uniformity result. In Section 4 we apply results of Donsker and Varadhan [5] in order to prove the asymptotics of (1). In Section 5 we prove the asymptotics of (2) by applying a large deviation for the BM on the torus. Applying elementary Sobolev space techniques we prove the solution of the variational problem in Section 6.

We conclude our introduction by stating and proving a technical lemma which will frequently be used. It is our essential tool for evaluating exponential moments of functionals of the Wiener process. The idea for its proof was communicated by S. R. S. Varadhan.

TECHNICAL LEMMA *Let $\{g_n\}_{n \in \mathbb{N}}$ be a sequence of non-negative measurable functions $g_n: \mathbb{R}^d \rightarrow [0, \infty]$ and $t > 0$ such that*

$$\lim_{n \rightarrow \infty} \sup_{x \in \mathbb{R}^d} E^{P_x} \left\{ \int_0^t g_n(X_s) ds \right\} = 0$$

where P_x is the Wiener process on $\mathbb{R}^d$ starting at $x \in \mathbb{R}^d$. Then it holds for all $c > 0$:

$$\lim_{n \rightarrow \infty} \sup_{x \in \mathbb{R}^d} \log E^{P_x} \exp \left\{ c \int_0^t g_n(X_s) ds \right\} = 0.$$

Proof The lower estimate is a straight application of Jensen's inequality and the hypothesis in the statement of the lemma. In order to get the upper estimate we chose N big enough such that

$$c \sup_{x \in \mathbb{R}^d} E^{P_x} \left\{ \int_0^t g_n(X_s) ds \right\} < 1 \quad \text{for all } n \geq N.$$

Now, we are able to go through the following calculation:

$$E^{P_0} \exp \left\{ c \int_0^t g_n(X_s) ds \right\}$$

$$\begin{aligned}
&= \sum_{k=0}^{\infty} \frac{c^k}{k!} E^{P_0} \left\{ \int_0^t g_n(X_s) ds \right\}^k \\
&= \sum_{k=0}^{\infty} c^k \int_{\{0 \leq s_1 \leq \dots \leq s_k \leq t\}} E^{P_0} \prod_{m=1}^k g_n(X(s_m)) ds_1 \dots ds_k
\end{aligned}$$

using conditional expectations we get

$$\begin{aligned}
&= \sum_{k=0}^{\infty} c^k \int_{\{0 \leq s_1 \leq \dots \leq s_k \leq t\}} E^{P_0} \left[E \left(\prod_{m=1}^k g_n(X(s_m)) \middle| \mathbb{F}_{s_{k-1}}^0 \right) \right] ds_1 \dots ds_k \\
&= \sum_{k=0}^{\infty} c^k \int_{\{0 \leq s_1 \leq \dots \leq s_k \leq t\}} E^{P_0} \left[\prod_{m=1}^{k-1} g_n(X(s_m)) E(g_n(X(s_k)) \middle| \mathbb{F}_{s_{k-1}}^0) \right] ds_1 \dots ds_k
\end{aligned}$$

where $\mathbb{F}_s = \sigma\{Xu; 0 \leq u \leq s\}$.

Using the Markov property and applying the non-negativeness of the functions g_n we find

$$\leq \sum_{k=0}^{\infty} c^k \sup_{x \in \mathbb{R}^d} E^{P_x} \left\{ \int_0^t g_n(X_s) ds \right\} \int_{\{0 \leq s_1 \leq \dots \leq s_{k-1} \leq t\}} E^{P_0} \left[\prod_{m=1}^{k-1} g_n(X(s_m)) \right] ds_1 \dots ds_{k-1}.$$

Now we repeat the last three steps and arrive at

$$\begin{aligned}
&\leq \sum_{k=0}^{\infty} c^k \left[\sup_{x \in \mathbb{R}^d} E^{P_x} \left\{ \int_0^t g_n(X_s) ds \right\} \right]^k \\
&\leq \frac{1}{1 - c \sup_{x \in \mathbb{R}^d} E^{P_x} \left\{ \int_0^t g_n(X_s) ds \right\}} \quad \text{for } n > N.
\end{aligned}$$

The last step is an application of the geometrical series and our special choice of N . Now, the Technical Lemma follows by letting $n \rightarrow \infty$. ■

2. DEFINITION OF THE DIRAC POLARON AND A FIRST APPROXIMATION LEMMA

Let $\{l_t\}_{t>0}$ be the Brownian local time as defined in Rogers and Williams [13, IV.43.3]. We will introduce the following notation:

For every $\lambda > 0$ let $\{l_{\lambda,t}\}_{t>0}$ be given by $l_{\lambda,t} = \int_0^t \lambda e^{-\lambda s} dl_s$.

Remark 2.1 Let $f(x) = \max\{0, 1 - |x|\}$ for $x \in \mathbb{R}$. Using f we define the functions

$$f^n(x) = n f(nx) \quad \text{for } n \in \mathbb{N};$$

$$\hat{f}^n(x) = 2 \int_{-\infty}^x f^n(y) dy - 1;$$

$$F^n(x) = \int_{\min\{x, 0\}}^{\max\{x, 0\}} |\hat{f}^n(y)| dy.$$

Following the lines of the proof of Theorem (IV.43.3) in Rogers and Williams [13] we get

THEOREM 2.2 (Tanaka's formula) *Let $\{X_t\}_{t \geq 0}$ be the Brownian motion starting at 0, and $\lambda > 0$. Then $\{l_{\lambda, t}\}_{t \geq 0}$ is a continuous increasing adapted process with:*

$$\lambda e^{-\lambda t} |X_t| = - \int_0^t \lambda^2 e^{-\lambda s} |X_s| ds + \int_0^t \lambda e^{-\lambda s} \operatorname{sgn}(X_s) dX_s + l_{\lambda, t}, \quad (2.2)$$

where sgn is defined as $\operatorname{sgn}(0) = 0$, $\operatorname{sgn}(x) = -1$ for $x < 0$ and $\operatorname{sgn}(x) = 1$ for $x > 0$.

As a consequence of the techniques applied in the proof of (IV.43.3) in Rogers and Williams [13] we get

Remark 2.3 An approximation of $l_{\lambda, t}$ is given by $C_{n, \lambda, t}(\omega) = \int_0^t \lambda e^{-\lambda s} f^n(X_s) ds$, where

$$C_{n, \lambda, t} = \lambda e^{-\lambda t} F^n(X_t) + \int_0^t \lambda^2 e^{-\lambda s} F^n(X_s) ds - \int_0^t \lambda e^{-\lambda s} \hat{f}^n(X_s) dX_s. \quad (2.3)$$

This approximation is uniformly over $[0, p[$ in the L^2 -sense and “almost surely” by passing to a subsequence if necessary.

Remark and Definition of the Dirac Polaron 2.4 Let ϑ_s be the shift operator by $\vartheta_s \omega(u) = \omega(u+s) - \omega(s)$. Then we define

$$\int_0^t \int_0^t \lambda \cdot e^{-\lambda|\sigma-s|} \delta(X_s - X_\sigma) ds d\sigma \quad \text{as} \quad 2 \int_0^t l_{\lambda, t-s}(\vartheta_s \omega) ds, \quad (2.4)$$

which according to Theorem 2.2 is explicitly given by

$$\begin{aligned} &= 2 \int_0^t \left[\lambda e^{-\lambda(t-s)} |X_t - X_s| + \int_0^{t-s} \lambda^2 e^{-\lambda u} |X_{s+u} - X_s| du \right. \\ &\quad \left. - \int_0^{t-s} \lambda e^{-\lambda u} \operatorname{sgn}(X_{s+u} - X_s) dX_{u+s} \right] ds. \end{aligned}$$

LEMMA 2.5 *There exists a function $n: [0, \infty[\rightarrow \mathbb{N}$ such that for all $q > 1$ and $\lambda > 0$*

$$\overline{\lim}_{t \rightarrow \infty} \frac{1}{qt} \log E^P \exp \left\{ \pm q \int_0^t \int_0^t \lambda e^{-\lambda|s-\sigma|} [\delta - f^{n(t)}](X_s - X_\sigma) ds d\sigma \right\} = 0.$$

Proof of Lemma 2.5 Using Remark 2.3 and Remark 2.4 we get:

$$\int_0^t \int_0^t \lambda e^{-\lambda|\sigma-s|} [\delta - f^{n(t)}](X_s - X_\sigma) ds d\sigma = 2 \int_0^t [l_{\lambda, t-s} - C_{n(t), \lambda, t-s}] ds$$

and

$$\left| \int_0^t [l_{\lambda, t-s} - C_{n(t), \lambda, t-s}] ds - \int_0^t \left[\int_0^{t-s} \lambda e^{-\lambda u} (\hat{f}^{n(t)}(X_{s+u} - X_s) - \operatorname{sgn}(X_{s+u} - X_s)) dX_{s+u} \right] ds \right| \leq (1 + \lambda t) \varepsilon_{n(t)},$$

where $\varepsilon_{n(t)} = \sup_{x \in \mathbb{R}} ||x| - F^{n(t)}(x)|$. If $\lim_{t \rightarrow \infty} n(t) = \infty$, then $\varepsilon_{n(t)} \downarrow 0$ if $t \uparrow \infty$. This reduces our problem to showing that

$$\overline{\lim}_{t \rightarrow \infty} \frac{1}{qt} \log E^P \exp \left\{ \pm 2q \int_0^t \left[\int_0^{t-s} \lambda e^{-\lambda u} (\hat{f}^{n(t)}(X_{s+u} - X_s) - \operatorname{sgn}(X_{s+u} - X_s)) dX_{s+u} \right] ds \right\} = 0.$$

Jensen's inequality yields:

$$\begin{aligned} E^P \exp \left\{ \pm 2q \int_0^t \left[\int_0^{t-s} \lambda e^{-\lambda u} (\hat{f}^{n(t)}(X_{s+u} - X_s) - \operatorname{sgn}(X_{s+u} - X_s)) dX_{s+u} \right] ds \right\} \\ \leq \frac{1}{t} \int_0^t E^P \exp \{ 2qt M_{t-s}^{n(t)} \} ds, \end{aligned}$$

where $M_v^{n(t)} = \pm \int_0^v \lambda e^{-\lambda u} (\hat{f}^{n(t)}(X_u) - \operatorname{sgn}(X_u)) dX_u$ is a P -martingale with quadratic variation process:

$$[M^{n(t)}]_v = \int_0^v \lambda^2 e^{-2\lambda u} (\hat{f}^{n(t)}(X_u) - \operatorname{sgn}(X_u))^2 du.$$

Using exponential martingale properties (IV.37.11) of Rogers and Williams [13] and Hölder's inequality, it follows that for all $v \in [0, t]$

$$\begin{aligned} E^P \exp \{ 2qt M_v^{n(t)} \} &= E^P \exp \{ 2qt M_v^{n(t)} - 4q^2 t^2 [M^{n(t)}]_v + 4q^2 t^2 [M^{n(t)}]_v \} \\ &\leq 1 \cdot \sqrt{E^P \exp \{ 8q^2 t^2 [M^{n(t)}]_v \}} \end{aligned}$$

$$\leq \sqrt{E^P \exp \{8q^2 t^2 [M^{n(t)}]_t\}}.$$

Lemma 2.5 will be shown if

$$\overline{\lim}_{t \rightarrow \infty} \frac{1}{2qt} \log E^P \exp \{8q^2 t^2 [M^{n(t)}]_t\} = 0.$$

Calculating the exponential moment as done in the “technical lemma” of Section 1 and using the notation $S(t) = \sup_{x \in \mathbb{R}} E^{P_x} [M^{n(t)}]_t$, we get

$$E^P \exp \{8q^2 t^2 [M^{n(t)}]_t\} \leq \sum_{k=0}^{\infty} [8q^2 t^2 S(t)]^k.$$

In order to prove the lemma, it suffices to show that $\lim_{t \rightarrow \infty} t^2 S(t) = 0$. But this follows from

$$\begin{aligned} E^{P_x} [M^{n(t)}]_t &= E^{P_x} \int_0^t \lambda^2 e^{-2\lambda s} [\hat{f}^{n(t)} - \text{sgn}]^2(X_s) ds \\ &\leq E^{P_x} \int_0^t \lambda^2 e^{-2\lambda s} 1_{\{|X(s)| < 1/n(t)\}} ds \\ &\leq \int_0^t \lambda^2 e^{-2\lambda s} P_x \left\{ |X_s| < \frac{1}{n(t)} \right\} ds. \end{aligned}$$

Using Lebesgue’s dominated convergence theorem we see that we can choose a function $n(t)$ such that $\lim_{t \rightarrow \infty} t^2 S(t) = 0$.

This proves Lemma 2.5. $\blacksquare$

3. THE UNIFORM APPROXIMATION RESULT

The aim of this section is to prove

THEOREM 3.1 *For $\varepsilon > 0$ and $q > 1$ there exists an $M(\varepsilon, q)$ such that:*

$$\sup_{n, m > M(\varepsilon, q)} \sup_{\lambda \in]0, 1[}$$

$$\overline{\lim}_{t \rightarrow \infty} \frac{1}{qt} \log E^P \exp \left\{ \pm q \int_0^t \int_0^t \lambda e^{-\lambda|\sigma-s|} [f^m - f^n](X_\sigma - X_s) d\sigma ds \right\} < \varepsilon.$$

The proof of Theorem 3.1 needs several preparatory lemmata as well as some new notation. For $n, m, r, s \in \mathbb{N}$ we write

$$I_{n,m}(r,s) = I(r,s) = \int_{(r-1)/\lambda}^{r/\lambda} \int_{(s-1)/\lambda}^{s/\lambda} \lambda e^{-\lambda|u-v|} [f^n - f^m](X_u - X_v) du dv.$$

For $\lambda > 0$, $t > 0$ we write $t_\lambda = \max\{n \in \mathbb{N}, n\lambda^{-1} \leq t\}$.

LEMMA 3.2 For fixed parameters $c > 1$, $\lambda \in]0, 1]$ and $t > 0$ we can find positive constants c_1 and c_2 such that:

$$\begin{aligned} & \frac{1}{ct} \log E^P \exp \left\{ c \int_0^{t_\lambda/\lambda} \int_0^{t_\lambda/\lambda} \lambda e^{-\lambda|\sigma-\delta|} [f^m - f^n](X_\sigma - X_s) d\sigma ds \right\} \\ & \leq \frac{c_1}{c} \lambda \log E^P \exp \{2cI(1, 1)\} + \frac{c_2}{c} \lambda \log \sup_{s \in \mathbb{N}} E^P \exp \{2c2^{(s+1)}(s+1)I(1, s+1)\}. \end{aligned}$$

Proof of Lemma 3.2:

$$\begin{aligned} & \log E^P \exp \left\{ c \int_0^{t_\lambda/\lambda} \int_0^{t_\lambda/\lambda} \lambda e^{-\lambda|\sigma-s|} [f^n - f^m](X_\sigma - X_s) d\sigma ds \right\} \\ & = \log E^P \exp \left\{ c \sum_{r=1}^{t_\lambda} I(r, r) + 2c \sum_{r=1}^{t_\lambda-1} I(r, r+1) + \dots + 2cI(1, t_\lambda) \right\} \\ & \leq \frac{t_\lambda}{2} \log E^P \exp \{2cI(1, 1)\} \\ & \quad + t_\lambda \sum_{r=1}^{t_\lambda-1} 2^{-(r+1)}(r+1)^{-2} \log E^P \exp \{2c(r+1)2^{(r+1)}I(1, r+1)\}. \end{aligned}$$

For the last step we used Hölder's inequality with $p=q=1/2$ several times to split the sum in the exponent. In order to handle expressions like $E^P \exp \{\sum_{r=1}^{t_\lambda-k} I(r, r+k)\}$ we observe that $I(r, r+k)$ and $I(r', r'+k)$ are independent if $r' > r+k$. This gives at most $(k+1)$ blocks of $t_\lambda(k+1)^{-1}$ independent I -variables:

$$\left. \begin{aligned} & I(1, 1+k), I(k+2, 2k+2), I(2k+3, 3k+3), \dots \\ & I(2, 2+k), I(k+3, 2k+3), I(2k+4, 3k+4), \dots \\ & \vdots \\ & I(k+1, 2k+1), I(2k+2, 3k+2), I(3k+3, 4k+3), \dots \end{aligned} \right\} (k+1) \text{ blocks.}$$

Using independence between members of a block, the generalized Hölder inequality (13.26) of Hewitt and Stromberg [10] to split the blocks and the fact, that $I(1, 1+r)$ are equal in law, we arrive at the estimate given above. Finally

observe that $\frac{1}{2}\lambda \leq t_\lambda/t \leq \lambda$ and that the series $\sum_{r=1}^{\infty} 2^{-(r+1)}(r+1)^{-2}$ converges. This proves Lemma 3.2. ■

LEMMA 3.3 For $\varepsilon \in]0, 1]$ it holds:

$$\lim_{N \rightarrow \infty} \sup_{n, m > N} \sup_{\lambda \in [\varepsilon, 1]} E^P \exp\{cI(1, 1)\} = 1.$$

Proof of Lemma 3.3 Using the notation introduced in Remark 2.3 we get:

$$\begin{aligned} & E^P \exp\{cI(1, 1)\} \\ &= E^P \exp\left\{2c \int_0^{1/\lambda} [C_{n, \lambda, \lambda-1-s} - C_{m, \lambda, \lambda-1-s}](\partial_s \omega) ds\right\} \\ &\leq \exp\{2c\delta_{n,m}\} E^P \exp\left\{2c \int_0^{1/\lambda} \int_0^{1/\lambda-s} \lambda e^{-\lambda u} [\hat{f}^n - \hat{f}^m](X_{u+s} - X_s) dX_{u+s} ds\right\}, \end{aligned}$$

where we took advantage of the uniform convergence of the first two summands in the definition of $C_{n, \lambda, \lambda-1-s}$. We also have $\delta_{n,m} \rightarrow 0$ for $n, m > N$ and $N \rightarrow \infty$ independent of λ . Following the argumentation of Lemma 2.5 we have:

$$E^P \exp\{cI(1, 1)\} \leq \exp\{2c\delta_{n,m}\} E^P \exp\{4c^2[M^{n,m}]_{\varepsilon^{-1}}\},$$

where

$$M_s^{n,m} = \int_0^s e^{-\lambda u} [\hat{f}^n - \hat{f}^m](X_u) dX_u$$

is a P -martingale with quadratic process

$$[M^{n,m}]_s = \int_0^s e^{-2\lambda u} [\hat{f}^n - \hat{f}^m](X_u) du.$$

Using Lebesgue's dominated convergence theorem it is easily seen that

$$\lim_{N \rightarrow \infty} \sup_{n, m > N} \sup_{x \in \mathbb{R}} E^{P_x} \left[\int_0^{\varepsilon^{-1}} [\hat{f}^n - \hat{f}^m](X_s) ds \right] = 0.$$

Now we can use the “technical lemma” of Section 1 to evaluate the exponential moment and we arrive at

$$\exp\{-2c\delta_{n,m}\} \leq E^P \exp\{cI(1, 1)\} \leq \exp\{2c\delta_{n,m}\}(1 + \varepsilon_{n,m}),$$

where

$$\lim_{N \rightarrow \infty} \sup_{n, m > N} \max \{ \delta_{n, m}; \varepsilon_{n, m} \} = 0 \quad \text{and} \quad \delta_{n, m}, \varepsilon_{n, m} \geq 0.$$

We also used Jensen's inequality in order to establish for the lower estimate the relation

$$1 \leq E^P \exp \left\{ 2c \int_0^{1/\lambda} \int_0^{1/\lambda - s} \lambda e^{-\lambda u} [\hat{f}^n - \hat{f}^m](X_{n+s} - X_s) dX_{n+s} ds \right\}.$$

This proves Lemma 3.3. $\blacksquare$

LEMMA 3.4 For $\varepsilon > 0$ and $c > 1$ it holds:

$$\lim_{N \rightarrow \infty} \sup_{n, m > N} \sup_{\lambda \in [\varepsilon, 1]} \sup_{s \in \mathbb{N}_0} E^P \exp \{ c(s+1)2^{(s+1)} I(1, s+1) \} = 1.$$

Proof of Lemma 3.4

$$\begin{aligned} & E^P \exp \{ c(s+1)2^{(s+1)} I(1, s+1) \} \\ &= E^P \exp \left\{ c(s+1)2^{(s+1)} \int_0^{\lambda^{-1}} \int_{(s-1)/\lambda}^{s/\lambda - v} \lambda e^{-\lambda|u-v|} [f^n - f^m](X_u - X_v) du dv \right\} \end{aligned}$$

using Jensen's inequality and the Markov property of BM we estimate

$$\leq \lambda \int_0^{\lambda^{-1}} E^P \exp \left\{ c(s+1)2^{(s+1)} \int_{(s-1)/\lambda - v}^{s/\lambda - v} e^{-\lambda u} [f^n - f^m](X_u) du \right\} dv.$$

Now we just look at what happens in the time interval $[(s-1)/\lambda - v, s/\lambda - v]$ and use again the Markov property. We get:

$$\leq \lambda \int_0^{\lambda^{-1}} \sup_{x \in \mathbb{R}} E^{P_x} \exp \left\{ c(s+1)2^{(s+1)} e^{\lambda v} e^{-(s-1)} \int_0^{\lambda^{-1}} e^{-\lambda u} [f^n - f^m](X_u) du \right\} dv.$$

We introduce $\beta(s, v, \lambda) = (s+1)2^{(s+1)} e^{\lambda v} e^{-(s-1)}$ and see

$$0 \leq \sup_{s \in \mathbb{N}} \sup_{\lambda \in [0, 1]} \sup_{v \in [0, \lambda^{-1}]} \beta(s, v, \lambda) \leq \beta < \infty.$$

Using Remark 2.3 and writing

$$\int_0^{\lambda^{-1}} e^{-\lambda u} [f^n - f^m](X_u) du = e^{-1} [F^n - F^m](X_{\lambda^{-1}})$$

$$+ \int_0^{\lambda^{-1}} \lambda e^{-\lambda u} [F^n - F^m](X_u) du - \int_0^{\lambda^{-1}} e^{-\lambda u} [\hat{f}^m - \hat{f}^n](X_u) dXu$$

we arrive at

$$\begin{aligned} & E^P \exp \{c(s+1)2^{(s+1)}I(1, s+1)\} \\ & \leq \exp \{c\beta\delta_{n,m}\} \lambda \int_0^{\lambda^{-1}} \sup_{x \in \mathbb{R}} E^{P^x} \exp \left\{ c\beta(s, v, \lambda) \int_0^{\lambda^{-1}} e^{-\lambda u} [\hat{f}^m - \hat{f}^n](X_u) dXu \right\} dv. \end{aligned}$$

Evaluating the exponential moment by using the “technical lemma”, we see that independent of $\lambda \in [\varepsilon, 1]$ we can find for $\gamma > 0$ a $N(\gamma) \in \mathbb{N}$ such that for $n, m > N(\gamma)$

$$\sup_{s \in \mathbb{N}} E^P \exp \{c(s+1)2^{(s+1)}I(1, s+1)\} \leq 1 + \gamma.$$

The lower estimate follows again by applying Jensen’s inequality which gives

$$\begin{aligned} & e^P \exp \{c(s+1)2^{(s+1)}I(1, s+1)\} \\ & \geq \exp \left\{ E^P \left[\lambda \int_0^{\lambda^{-1}} c\beta(s, v, \lambda) E^{P^{x_v}} \left\{ \int_{(s-1)/\lambda - v}^{s/\lambda - v} e^{-\lambda u} [f^n - f^m](X_u) du \right\} dv \right] \right\}. \end{aligned}$$

Using (3.4.1) and recalling the uniform convergence of the sequence $\{F^n\}_{n \in \mathbb{N}}$ we arrive at

$$E^P \exp \{c(s+1)2^{(s+1)}I(1, s+1)\} \geq \exp \{-c\beta\delta_{n,m}\}$$

where

$$\delta_{n,m} = \|F^n - F^m\| \quad \text{and} \quad \lim_{N \rightarrow \infty} \sup_{n, m > N} \delta_{n,m} = 0.$$

This proves Lemma 3.4. ■

LEMMA 3.5 For $\delta > 0$ but close to 0, and $\lambda > 0$ we can find a $N = N(\delta, \gamma) \in \mathbb{N}$ such that for all $\lambda \in]0, \delta]$

$$\sup_{t \in [\delta^{-1}, \lambda^{-1}]} \sup_{n, m > N} \sup_{x \in \mathbb{R}} E^P \exp \left\{ ct \int_{\mathbb{R}} [f^n - f^m](y-x) l_t^\lambda(y-x) dy \right\} \leq \exp \{c\lambda^{-1}2\gamma\} + \gamma.$$

In Lemma 3.5 we used the notation $l_t^\lambda(y) = t^{-1} \int_0^t e^{-\lambda u} \delta(X(u) - y) du$.

First, don’t worry about the missing λ in front of $e^{-\lambda u}$! We think of l_t^λ as a “local time” and have to study its behavior over intervals $I_x = \sup \{f^n(\cdot - x) - f^m(\cdot - x)\}$ of length $2M$ where $M^{-1} = \min\{n, m\}$. Actually we will study the behavior of l_t^λ over small intervals and therefore think of l_t^λ as of the form $l_t^\lambda(y) = c + s_x(y)$ for $y \in I_x$, where s_x is a random perturbation of c , the “mean” of the “local time” over the small interval I_x . Looking at how big this perturbation can be, Lemma 3.5 will be shown if we prove the existence of a $N(\delta, \gamma)$ such that for δ close to 0 and all $\lambda \in]0, \delta]$

$$\sup_{\lambda \in]0, \delta]} \sup_{t \in [\delta^{-1}, \lambda^{-1}]} \sup_{n, m > N} \sup_{x \in \mathbb{R}} \sum_{k=1}^{\infty} \exp\{ct2(\gamma+k)\} P\left\{\sup_{|x-y| \leq M} |l_t^\lambda(x) - l_t^\lambda(y)| > \gamma+k-1\right\} < \gamma. \quad (3.4.2)$$

In order to prove (3.4.2) we apply a technique developed in Donsker and Varadhan [7, Section 3]. It is not difficult to get the needed modifications of the lemmata from Donsker and Varadhan [7, Section 3] in order to prove (3.4.2).

Proof of (3.4.2) Using the scaling properties of Brownian motion we observe the following "equality in law":

$$\begin{aligned} l_t^\lambda(\omega, y) &= \frac{1}{t} \int_0^t e^{-\lambda u} \delta(X(u) - y) du \\ &= \int_0^1 e^{-\lambda ts} \delta[\sqrt{t}(X(s) - y/\sqrt{t})] ds = \frac{1}{\sqrt{t}} l_1^{\lambda t}(\omega, y/\sqrt{t}) \end{aligned}$$

where $\lambda t \in [0, 1]$. We write $\alpha = \lambda t$.

Therefore:

$$P\left\{\sup_{|x-y| \leq M} |l_t^\lambda(\omega, x) - l_t^\lambda(\omega, y)| \geq a\right\} = P\left\{\sup_{|x-y| \leq M/\sqrt{t}} |l_1^\alpha(\omega, x) - l_1^\alpha(\omega, y)| \geq a\sqrt{t}\right\}.$$

Now we apply Lemma 3.4 of Donsker and Varadhan [7]. Let $\psi(x) = \exp\{c_1|x|^\rho\} - 1$, $p(u) = |u|^{9/2}$ and $f(x) = l_1^\alpha(\omega, x)$, where c_1 , ρ and 9 are as in the modified Lemma 3.3 of Donsker and Varadhan [7]. We note that $\psi(x)$ and $p(u)$ satisfy the hypothesis of Lemma 3.4. Moreover if we pick x and y such that $|x-y| \leq \delta$, where δ is small enough then the integral hypothesis of Lemma 3.4 for this ω , p , f follows from Lemma 3.3. We conclude from Lemma 3.4:

$$\sup_{|x-y| \leq M/\sqrt{t}} |l_1^\alpha(\omega, x) - l_1^\alpha(\omega, y)| \leq 8 \int_0^{M/\sqrt{t}} \psi^{-1}\left(\frac{4M_\alpha(\omega)}{u^2}\right) dp(u)$$

where

$$M_\alpha(\omega) = \iint_{|y-x| \leq 1} \left[\exp\left\{c_1 \left(\frac{|l_1^\alpha(\omega, x) - l_1^\alpha(\omega, y)|}{|x-y|^{9/2}}\right)^\rho\right\} - 1 \right] dx dy.$$

Using $\psi^{-1}(x) = [(1/c_1) \log(1+x)]^{1/\rho}$ as well as the explicit form of p we get:

$$\sup_{|x-y| \leq M/\sqrt{t}} |l_1^\alpha(\omega, x) - l_1^\alpha(\omega, y)| \leq \frac{49}{c_1^{1/\rho}} \int_0^{M/\sqrt{t}} \left[\log\left(1 + \frac{4M_\alpha(\omega)}{u^2}\right) \right]^{1/\rho} du.$$

Introducing $c = \rho/\sqrt{c_1}/49$ we have

$$P \left\{ \sup_{|x-y| \leq M/\sqrt{t}} |l_1^x(\omega, x) - l_1^x(\omega, y)| \geq a\sqrt{t} \right\} \\ \leq P \left\{ c^{-1} \int_0^{M/\sqrt{t}} \rho \sqrt{\log \left(1 + \frac{4M_x(\omega)t}{u^2} \right)} u^{(9/2)-1} du \geq a\sqrt{t} \right\}.$$

We are interested in the behavior of this probability for $t \in [\delta^{-1}, \lambda^{-1}]$ where $0 < \lambda < \delta$ and δ close to 0 as well as $M = [\min\{n, m\}]^{-1}$ close to 0. Therefore we are in the position to apply the fundamental theorem of calculus:

$$\leq P \left\{ c^{-1} \left(\frac{M}{\sqrt{t}} \right)^{9/2} \sqrt{\log \left(1 + \frac{4M_x(\omega)t}{M^2} \right)} \geq a\sqrt{t} \right\} \\ \leq P \left\{ \log \left(1 + \frac{4M_x(\omega)t}{M^2} \right) \geq c^\rho a^\rho M^{-\rho 9/2} t \right\}$$

where we used the identity $(9/4) + \frac{1}{2} \rho = 1$. Writing $c^* = c^\rho$ we get

$$\leq P \left\{ M_x(\omega) \geq \frac{M^2}{4t} \exp \{ c^* a^\rho M^{-\rho 9/2} t \} - \frac{M^2}{4t} \right\}.$$

Using the fact that $M^2/4t$ is close to 0 we estimate

$$\leq P \left\{ M_x(\omega) \geq \frac{M^2}{8t} \exp \{ c^* a^\rho M^{-\rho 9/2} t \} \right\}$$

and applying the Chebejev-inequality we arrive at

$$\leq \frac{8t}{M^2} \exp \{ -c^* a^\rho M^{-\rho 9/2} t \} k^*$$

where we also used the result of Lemma 3.3 of Donsker and Varadhan [7] that

$$\sup_{\alpha \in]0, 1]} EM_\alpha = k^* < \infty.$$

Now we are in the position of going through the following estimation:

$$\sum_{k=1}^{\infty} \exp \{ ct^2(\gamma + k) \} P \left\{ \sup_{|x-y| \leq M} |l_t^x(x) - l_t^x(y)| > \gamma + k - 1 \right\}$$

$$\begin{aligned} &\leq \frac{8t}{M^2} k^* \sum_{k=1}^{\infty} \exp\{2ct(\gamma+k) - c^*(\gamma+k-1)^\rho M^{-\rho\beta/2}t\} \\ &\leq \frac{8t}{M^2} k^* \sum_{k=1}^{\infty} \exp\{-t(\gamma+k-1)^\rho c_M\} \end{aligned}$$

where

$$c_M = \sup_{k \in \mathbb{N}} \left[c^* M^{-\rho\beta/2} - 2c \frac{\gamma+k}{(\gamma+k-1)^\rho} \right]$$

and we see that $c_M \uparrow \infty$ for $M \downarrow 0$. And we get right away

$$\leq \frac{8t}{M^2} k \exp\{-tc_M \gamma^\rho\} + \frac{8t}{M^2} k^* \int_{\gamma}^{\infty} \exp\{-tw^\rho c_M\} dw$$

and an obvious substitution gives

$$\leq \frac{8t}{M^2} k \exp\{-tc_M \gamma^\rho\} + \frac{8k^* \gamma^{1-\rho}}{M^2 c_M \rho} \exp\{-\delta^{-1} \gamma^\rho c_M\}.$$

Recalling the definitions $\beta \in]0, 1]$, $\beta = 1 - \beta/2$ and $\rho = 2/(2 - \beta)$ we see that it is possible to find for $c > 1$, $\gamma > 0$, $\delta > 0$ a $N(\gamma)$ such that for all $\lambda \in]0, \delta]$:

$$\sup_{n, m > N} \sup_{t \in [\delta^{-1}, \lambda^{-1}]} \sum_{k=1}^{\infty} \exp\{ct2(\gamma+k)\} P \left\{ \sup_{|x-y| \leq M/\sqrt{t}} |l_1^k(\omega, x) - l_1^k(\omega, y)| > \gamma + k - 1 \right\} < \gamma.$$

This proves (3.4.2) and Lemma 3.5, too. ■

Proof of Theorem 3.1 It is obvious that the $\pm$ in the statement of the theorem doesn't matter and that it is enough to consider the $+$ sign. Using the notation $t_\lambda = \max\{n \in \mathbb{N}; n\lambda^{-1} \leq t\}$, $t^* = t_\lambda \lambda^{-1}$ and applying Hölder's inequality for $p=q=1/2$ we get by taking advantage of the independence and stationarity of not overlapping Brownian increments that

$$\begin{aligned} &\frac{1}{qt} \log E^P \exp \left\{ q \int_0^t \int_0^t \lambda \exp[-\lambda|\sigma-s|] [f^m - f^n](X_\sigma - X_s) d\sigma ds \right\} \\ &\leq \frac{1}{2qt} \log E^P \exp \left\{ 2q \int_0^{t_\lambda \lambda^{-1}} \int_0^{t_\lambda \lambda^{-1}} \lambda \exp[-\lambda|\sigma-s|] [f^m - f^n](X_\sigma - X_s) d\sigma ds \right\} \end{aligned}$$

$$\begin{aligned}
 & + \frac{1}{2qt} \log E^P \exp \left\{ 2q \int_0^{t^*} \int_0^{t^*} \lambda \exp[-\lambda|\sigma-s|] [f^m - f^n](X_\sigma - X_s) d\sigma ds \right\} \\
 & + \frac{1}{2qt} \log E^P \exp \left\{ 4q \int_0^{t^*} \int_{t^*}^t \lambda \exp[-\lambda|\sigma-s|] [f^m - f^n](X_\sigma - X_s) d\sigma ds \right\}.
 \end{aligned}$$

Now we write

$$\begin{aligned}
 & \sup_{\lambda \in]0, 1]} \overline{\lim}_{t \rightarrow \infty} \frac{1}{qt} \log E^P \exp \left\{ q \int_0^t \int_0^t \lambda \exp[-\lambda|\sigma-s|] [f^m - f^n](X_\sigma - X_s) d\sigma ds \right\} \\
 & \leq \sum_{i=1}^3 \max \{ S_{]0, \delta]}^i, S_{[\delta, 1]}^i \}
 \end{aligned}$$

where $S_{]0, \delta]}^i$ is the supremum over $\lambda \in]0, \delta]$ of the i th summand in the sum above and where $S_{[\delta, 1]}^i$ is the supremum over $\lambda \in [\delta, 1]$ of the i th summand in the sum above. The analysis of $S_{[\delta, 1]}^1$ is possible by using Lemmas (3.2), (3.3) and (3.4). For every $q > 1$, $\delta > 0$ and $\gamma > 0$ we find a $M_1 = M_1(\delta, \gamma, q)$ such that for $n, m > M_1$ it holds: $-\lambda/6 \leq S_{[\delta, 1]}^1 \leq \gamma/6$. Applying the techniques used in the proof of Lemma 3.3 directly to $S_{[\delta, 1]}^2$ gives us the existence of a $M_2 = M_2(\delta, \gamma, q)$ such that for $n, m > M_2$ we get: $-\gamma/6 \leq S_{[\delta, 1]}^2 \leq \gamma/6$. Likewise we can apply the techniques used in the proof of Lemma 3.4 to handle the $S_{[\delta, 1]}^3$ -term. In order to do this we split the third summand in t_λ parts by using Hölder's inequality and we get expressions fitting the statement of Lemma 3.4. This way we find a $M_3 = M_3(\delta, \gamma, q)$ such that for $n, m > M_3$ we find: $-\gamma/6 \leq S_{[\delta, 1]}^3 \leq \gamma/6$.

In order to prove the uniform behavior for small λ we will give an explicit treatment of the case $S_{]0, \delta]}^1$. The cases $S_{]0, \delta]}^2$ and $S_{]0, \delta]}^3$ follow by obvious modifications.

Using Lemma 3.2 we see:

$$\begin{aligned}
 & \sup_{\lambda \in]0, \delta]} \overline{\lim}_{t \rightarrow \infty} \frac{1}{2qt} \log E^P \exp \left\{ 2q \int_0^{t_\lambda/\lambda} \int_0^{t_\lambda/\lambda} \lambda \exp[-\lambda|\sigma-s|] [f^m - f^n](X_\sigma - X_s) d\sigma ds \right\} \\
 & \leq \sup_{\lambda \in]0, \delta]} \frac{\lambda}{2q} c_1(\lambda) \log E^P \exp \{ 4qI(1, 1) \} \\
 & \quad + \sup_{\lambda \in]0, \delta]} \frac{\hat{\lambda}}{2q} c_2(\lambda) \log \sup_{n \in \mathbb{N}} E^P \exp \{ 4q(n+1)2^{(n+1)}I(1, n+1) \}.
 \end{aligned}$$

Applying Jensen's inequality we have seen in Lemma 3.3 that

$$\begin{aligned}
E^P \exp \{4qI(1, 1)\} &\leq \lambda \int_0^{\lambda^{-1}} E^P \exp \left\{ 4q \int_0^{\lambda^{-1}-s} e^{-\lambda u} [f^n - f^m](X_u) du \right\} ds \\
&\leq \lambda \int_0^{\lambda^{-1}-\delta^{-1}} I_\lambda(s) ds + \lambda \int_{\lambda^{-1}-\delta^{-1}}^{\lambda^{-1}} I_\lambda(s) ds
\end{aligned}$$

where

$$I_\lambda(s) = E^P \exp \left\{ 4q \int_0^{\lambda^{-1}-s} e^{-\lambda u} [f^n - f^m](X_u) du \right\}.$$

Remember that $\lambda \in]0, \delta]$ and δ close to 0. Therefore we observe that in the second integral $\lambda^{-1} - s \leq \delta^{-1}$ and applying arguments used in the proof of Lemma 3.3 gives us a $M_4^* = M_4^*(\delta, \gamma, q)$ such that for $n, m > M_4^*$ $1 - \gamma I(s) \leq 1 + \gamma$ for $s \in [\lambda^{-1} - \delta^{-1}, \lambda^{-1}]$. Studying the first integral we see that the integrant can be written in the form

$$I_\lambda(s) = E^P \exp \left\{ 4qt \int_{\mathbb{R}} [f^n - f^m](x) l_t^\lambda(x) dx \right\} \quad \text{where } t = \lambda^{-1} - s$$

where l_t^λ was defined in Lemma 3.5.

Applying Lemma 3.5 we find for given $\delta > 0$, $\gamma > 0$, $q > 1$ a $M_4^{**} = M_4^{**}(\delta, \gamma, q)$ such that $n, m > M_4^{**}$ implies

$$\sup_{t \in [\delta^{-1}, \lambda^{-1}]} E^P \exp \left\{ 4qt \int_{\mathbb{R}} [f^n - f^m](x) l_t^\lambda(x) dx \right\} \leq \exp \{4q\lambda^{-1}2\gamma\} + \gamma.$$

For $n, m > \max\{M_4^*, M_4^{**}\}$ we arrive at

$$\begin{aligned}
&\sup_{\lambda \in]0, \delta]} \frac{\lambda}{2q} c_1(\lambda) \log E^P \exp \{4qI(1, 1)\} \\
&\leq \sup_{\lambda \in]0, \delta]} \frac{\lambda}{2q} c_1(\lambda) \log [\lambda(\lambda^{-1} - \delta^{-1}) \{ \exp(8q\lambda^{-1}) + \gamma \} + \lambda\delta^{-1}(1 + \gamma)] \\
&\lambda \leq \sup_{\lambda \in]0, \delta]} \frac{\lambda}{2q} c_1(\lambda) \log [\{u_p(8q\gamma\lambda^{-1})\} + 1]
\end{aligned}$$

and using the inequality $\log(x+1) \leq x$ we arrive at

$$\leq 4c_1(\lambda)\gamma.$$

In order to study

$$\sup_{\lambda \in]0, \delta]} \frac{\lambda}{2q} c_2(\lambda) \log \sup_{n \in \mathbb{N}} E^P \exp \{4q(n+1)2^{(n+1)}I(1, n+1)\}$$

Lemma 3.4 tells us to look at

$$\lambda \int_0^{\lambda^{-1}} \sup_{s \in \mathbb{N}} \sup_{x \in \mathbb{R}} E^{P_x} \exp \left\{ 4q\beta(s, v, \lambda) \int_0^{\lambda^{-1}} e^{-\lambda u} [f^m - f^n](X_u) du \right\} dv$$

where $\beta(s, v, \lambda) = (s+1)2^{(s+1)} e^{\lambda v} e^{-(s-1)}$ is uniformly bounded in all variables. This conducts to study expressions like

$$\begin{aligned} & \sup_{x \in \mathbb{R}} E^{P_x} \exp \left\{ 4qc \int_0^{\lambda^{-1}} e^{-\lambda u} [f^m - f^n](X_u) du \right\} \\ &= \sup_{x \in \mathbb{R}} E^{P_x} \exp \left\{ 4qc\lambda^{-1} \int_{\mathbb{R}} [f^m - f^n](x) l_{\lambda^{-1}}^1(x) dx \right\}. \end{aligned}$$

Observing that

$$\begin{aligned} & E^{P_x} \exp \left\{ 4qc\lambda^{-1} \int_{\mathbb{R}} [f^m - f^n](y) l_{\lambda^{-1}}^1(y) dy \right\} \\ &= E^P \exp \left\{ 4qc\lambda^{-1} \int_{\mathbb{R}} [f^m - f^n](y-x) l_{\lambda^{-1}}^1(y-x) dy \right\} \end{aligned}$$

we can apply Lemma 3.5 and get the needed result. We finally arrive at

$$\begin{aligned} & \sup_{\lambda \in]0, \delta]} \frac{\lambda}{2q} c_2(\lambda) \log \sup_{n \in \mathbb{N}} E^P \exp \{4q(n+1)2^{(n+1)}I(1, n+1)\} \\ & \times \sup_{\lambda \in]0, \delta]} \frac{\lambda}{2q} c_2(\lambda) \log \left\{ \sup_{n \in \mathbb{N}} \sup_{v \in [0, 1/\lambda]} \exp \{4q\beta(n, v, \lambda)\lambda^{-1}2\gamma\} + \gamma \right\} \leq 4\gamma\beta \end{aligned}$$

for $n, m > M$ and M large enough. We did not prove explicitly the lower estimates because they follow the same way as shown in the proof of Lemma 3.3 and Lemma 3.4. Now it is obvious how to treat the missing two cases $S_{[0, \delta]}^2$ and $S_{[0, \delta]}^3$. Finally we control the sum $\sum_{i=1}^3 \max\{S_{[0, \delta]}^i, S_{[\delta, 1]}^i\}$ and can determine for given $q > 1$ and $\gamma \leq 0$ a $M = M(q, \gamma)$ such that for $n, m > M$

$$\sup_{n, m > M} \sup_{\lambda \in]0, 1]} \overline{\lim}_{t \rightarrow \infty} \frac{1}{qt} \log E^P \exp \left\{ \pm q \int_0^t \int_0^t \lambda e^{-\lambda|\sigma-s|} [\hat{f}^m - \hat{f}^n](X_s - X_\sigma) ds d\sigma \right\} < \gamma.$$

This proves Theorem 3.1. $\blacksquare$

4. THE ASYMPTOTICS OF $N(\lambda, t)$

The goal of this section is to prove

THEOREM 4.1

$$\lim_{\lambda \rightarrow 0} \lim_{t \rightarrow \infty} \frac{1}{t} \log N(\lambda, t) = \sup_{\substack{\phi \in L^2(\mathbb{R}) \\ \|\phi\|_2 = 1}} \left\{ 2 \|\phi\|_4^4 - \frac{1}{2} \|\phi'\|_2^2 \right\}.$$

Our proof will rely on general results proved by Donsker and Varadhan [5] which we formulate in the following theorem. This theorem is a combination of Theorems 3.2 and 4.1, Lemmas 4.2 and 4.4 of Donsker and Varadhan [5].

THEOREM 4.2 *Let $f: \mathbb{R} \rightarrow \mathbb{R}^+$ be a bounded, continuous and positive definite function. P is the Brownian motion on $\mathbb{R}$ starting at 0. Then the following holds:*

$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \left\{ \int_0^t \int_0^t \lambda e^{-\lambda|\sigma-s|} f(X_s - X_\sigma) d\sigma ds \right\} \\ = \sup_{Q \in M_S(\Omega)} \left[2E^Q \int_0^\infty \lambda e^{-\lambda u} f(X_u - X_0) du - H(Q) \right] \end{aligned} \quad (4.2.1)$$

$$\begin{aligned} \lim_{\lambda \rightarrow 0} \lim_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \left\{ \int_0^t \int_0^t \lambda e^{-\lambda|\sigma-s|} f(X_s - X_\sigma) d\sigma ds \right\} \\ = \sup_{\substack{\phi \in L^2(\mathbb{R}) \\ \|\phi\|_2 = 1}} \left[2 \iint_{\mathbb{R} \times \mathbb{R}} \phi(x)^2 f(x-y) \phi(y)^2 dx dy - \frac{1}{2} \int_{\mathbb{R}} |\phi'(y)|^2 dy \right]. \end{aligned} \quad (4.2.2)$$

$M_S(\Omega)$ is the space of all stationary processes on $\Omega = C(\mathbb{R}, \mathbb{R})$. $H(Q)$ is the entropy-functional with respect to the BM as introduced in Donsker and Varadhan [4, Section 3].

First, we observe that the functions f and f^n , introduced in Remark 2.1, are positive definite. The Fourier transform of f is given by $\hat{f}(\xi) = (2/\xi^2)(1 - \cos(\xi))$. We see that because of $\hat{f} \in L^1(\mathbb{R})$ and of Hewitt and Stromberg [10, (21.49)] the inverse Fourier transform of $\hat{f}$ is f itself.

Using the approximation results obtained in Lemma 2.5 together with Theorem 3.1 we will prove the

PROPOSITION 4.3 For every real $p > 1$ holds

$$\overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log N(\lambda, t) \leq \overline{\lim}_{n \rightarrow \infty} \overline{\lim}_{t \rightarrow \infty} \frac{1}{pt} \log E^p \exp \left\{ p \int_0^t \int_0^t \lambda e^{-\lambda|s-\sigma|} f^n(X_s - X_\sigma) ds d\sigma \right\}, \quad (1)$$

and

$$\underline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log N(\lambda, t) \geq \overline{\lim}_{n \rightarrow \infty} \underline{\lim}_{t \rightarrow \infty} \frac{p}{t} \log E^p \exp \left\{ \frac{1}{p} \int_0^t \int_0^t \lambda e^{-\lambda|s-\sigma|} f^n(X_s - X_\sigma) ds d\sigma \right\}, \quad (2)$$

Proof of Proposition 4.3 We will show inequality (2) of Proposition 4.3. The proof of inequality (1) works similarly. Using the “lower estimate version” of Hölder’s inequality (13.6) of Hewitt and Stromberg [10] we get for arbitrary $\varepsilon > 0$:

$$\begin{aligned} & \underline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^p \exp \left\{ \int_0^t \int_0^t \lambda e^{-\lambda|s-\sigma|} \delta(X_s - X_\sigma) ds d\sigma \right\} \\ & \geq \underline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^p \exp \left\{ \int_0^t \int_0^t \lambda e^{-\lambda|s-\sigma|} [\delta - f^{n(t)} + f^{n(t)} - f^m + f^m](X_s - X_\sigma) ds d\sigma \right\} \\ & \geq \underline{\lim}_{t \rightarrow \infty} \frac{1}{t} p \log E^p \exp \left\{ \frac{1}{p} \int_0^t \int_0^t \lambda e^{-\lambda|s-\sigma|} f^m(X_s - X_\sigma) ds d\sigma \right\} \\ & \quad + \underline{\lim}_{t \rightarrow \infty} - \frac{p}{2qt} \log E^p \exp \left\{ - \frac{2q}{p} \int_0^t \int_0^t \lambda e^{-\lambda|s-\sigma|} [\delta - f^{n(t)}](X_s - X_\sigma) ds d\sigma \right\} \\ & \quad + \underline{\lim}_{t \rightarrow \infty} - \frac{p}{2qt} \log E^p \exp \left\{ - \frac{2q}{p} \int_0^t \int_0^t \lambda e^{-\lambda|s-\sigma|} [f^{n(t)} - f^m](X_s - X_\sigma) ds d\sigma \right\}. \end{aligned}$$

Using Lemma 2.5 and Theorem 3.1 we get for $m > M(\varepsilon, q)$

$$\underline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log N(\lambda, t) \geq \underline{\lim}_{t \rightarrow \infty} \frac{p}{t} \log E^p \exp \left\{ \frac{1}{p} \int_0^t \int_0^t \lambda e^{-\lambda|s-\sigma|} f^m(X_s - X_\sigma) ds d\sigma \right\} - \varepsilon$$

which proves inequality (2). ■

Proof of Theorem 4.1

Step 1 Using Proposition 4.3 we get for $p > 1$, $\varepsilon > 0$ and $m > M^*(\varepsilon, q)$

$$\lim_{\lambda \rightarrow 0} \lim_{t \rightarrow \infty} \frac{1}{t} \log N(\lambda, t) \geq p \lim_{\lambda \rightarrow 0} \lim_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \left\{ \frac{1}{p} \int_0^t \int_0^t \lambda e^{-\lambda|s-\sigma|} f^m(X_s - X_\sigma) ds d\sigma \right\} - \varepsilon.$$

Now we apply Lemma 4.4 of Donsker and Varadhan [5] and get

$$\geq p \sup_{\substack{\phi \in L^2(\mathbb{R}) \\ \|\phi\|_2 = 1}} \left[\frac{2}{p} \int_{\mathbb{R}} \int_{\mathbb{R}} \phi(x)^2 f^m(x-y) \phi(y)^2 dx dy - \frac{1}{2} \|\phi'\|_2^2 \right] - \varepsilon$$

for all $m > M^*(\varepsilon, q)$. Lacking a nice monotony argument we use Remark 6.2.7 of Section 6 and (21.37) of Hewitt and Stromberg [10] and get: For $\varepsilon > 0$ there exists a $M(\varepsilon)$ such that for $m > M(\varepsilon)$ and $\phi \in W^{2,1}(\mathbb{R})$ with $S_m(\phi) > S_m - 1$ it yields:

$$\left| \int_{\mathbb{R}} [\phi^2 * f^m(x) - \phi^2] \phi^2(x) dx \right| < \varepsilon c.$$

The constant c and the notation S_m are explained in Remark 6.2.7. Now for every $p > 1$ and $\varepsilon > 0$ we get:

$$\lim_{\lambda \rightarrow 0} \lim_{t \rightarrow \infty} \frac{1}{t} \log N(\lambda, t) \geq p \sup_{\substack{\phi \in L^2(\mathbb{R}) \\ \|\phi\|_2 = 1}} \left\{ \frac{2}{p} \|\phi\|_4^4 - \frac{1}{2} \|\phi'\|_2^2 \right\} - \varepsilon(1+2c).$$

Step 2 Using Proposition 4.3 we are able to find for every $\varepsilon > 0$ and $p > 0$ a $M(\varepsilon, q)$ such that

$$\overline{\lim}_{\lambda \rightarrow 0} \overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log N(\lambda, t) \leq \overline{\lim}_{\lambda \rightarrow 0} \overline{\lim}_{t \rightarrow \infty} \frac{1}{pt} \log E^P \exp \left\{ p \int_0^t \int_0^t \lambda e^{-\lambda|s-\sigma|} f^n(X_s - X_\sigma) ds d\sigma \right\} + \varepsilon$$

for all $n > M(\varepsilon, q)$. Theorem 4.2 gives

$$\leq \frac{1}{p} \sup_{\substack{\phi \in L^2(\mathbb{R}) \\ \|\phi\|_2 = 1}} \left[2p \int_{\mathbb{R}} \int_{\mathbb{R}} \phi(x)^2 f^n(x-y) \phi(y)^2 dx dy - \frac{1}{2} \|\phi'(y)\|_2^2 \right] + \varepsilon.$$

Now by using the notation $\hat{f}$ for the Fourier transform of f we find

$$\int_{\mathbb{R}} \int_{\mathbb{R}} \phi(x)^2 f^n(x-y) \phi(y)^2 dx dy = \langle \widehat{\phi^2}; \widehat{f^n * \phi^2} \rangle_{L^2} \uparrow \langle \widehat{\phi^2}; \widehat{\phi^2} \rangle_{L^2} = \|\phi\|_4^4.$$

This is a consequence of the fact that ϕ can be chosen as an element of $W^{2,1}(\mathbb{R})$. An imbedding-theorem (Adams [1, Corollary 5.6]) says that in our situation ϕ is also an element of $L^q(\mathbb{R})$ for $q \geq 2$. Especially $\phi \in L^4(\mathbb{R})$ or $\phi^2 \in L^2(\mathbb{R})$. Now making use of the fact that the Fourier transform is a L^2 -isometry and $f^n \uparrow 1$ we get

$$\leq \frac{1}{p} \sup_{\substack{\phi \in L^2(\mathbb{R}) \\ |\phi|_2 = 1}} \{2p|\phi|_4^4 - \frac{1}{2}|\phi'|_2^2\} + \varepsilon.$$

Step 3 We show that it is possible to take the limit $p \rightarrow 1$. Using Lemma 6.2 we see that for $p \in [0, \infty[$

$$S(p) = \sup_{\substack{\phi \in L^2(\mathbb{R}) \\ |\phi|_2 = 1}} \{2p|\phi|_4^4 - \frac{1}{2}|\phi'|_2^2\}$$

is a finite convex function and therefore continuous. Passing with $p \rightarrow 1$ and using the results of Step 1 and Step 2 we get

$$\lim_{\lambda \rightarrow 0} \lim_{t \rightarrow \infty} \frac{1}{t} \log N(\lambda, t) = \sup_{\substack{\phi \in L^2(\mathbb{R}) \\ |\phi|_2 = 1}} \{2|\phi|_4^4 - \frac{1}{2}|\phi'|_2^2\}.$$

This finishes the proof of Theorem 4.1. $\blacksquare$

Remark 4.5

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \left\{ \int_0^t \int_0^t \lambda e^{-\lambda|\sigma-s|} \delta(X_s - X_\sigma) d\sigma ds \right\} = \tilde{g}(\lambda)$$

exists and is finite for all $\lambda > 0$.

We define a sequence of finite continuous functions $\{\Gamma_n(p)\}_{n \in \mathbb{N}}$, $p \in [1-\varepsilon, 1+\varepsilon]$, convex in p , by

$$\Gamma_n(p) = \overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \left\{ p \int_0^t \int_0^t \lambda e^{-\lambda|\sigma-s|} f^n(X_s - X_\sigma) d\sigma ds \right\}.$$

We see that $\Gamma(p) = \limsup_{n \in \mathbb{N}} \Gamma_n(p)$ is again a convex function. Using Lemma 2.5 and Theorem 3.1 and going through the arguments we gave to prove (4.3.1) we see that $\Gamma(p)$ is finite for $p \neq 1$. This shows that Γ is finite on $[1-\varepsilon, 1+\varepsilon]$ and therefore continuous at $p=1$. This proves

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log N(\lambda, t) = \Gamma(1)$$

$$= \overline{\lim}_{n \rightarrow \infty} \overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \left\{ \int_0^t \int_0^t \lambda e^{-\lambda|\sigma-s|} f^n(X_s - X_\sigma) d\sigma ds \right\} < \infty.$$

And we found an expression describing the free energy of the Dirac polaron for finite λ .

5. THE MEANFIELD ASYMPTOTIC

In this section we prove

THEOREM 5.1 *Let P be the Wiener process on $\mathbb{R}$ starting at 0 then:*

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \left\{ \frac{2}{t} \int_0^t \int_0^t \delta(X_s - X_\sigma) ds d\sigma \right\} = \sup_{\substack{\phi \in L^2_{\mathbb{R}} \\ \|\phi\|_2 = 1}} [2|\phi|_4^4 - \frac{1}{2}|\phi'|_2^2].$$

We use the defining relation:

$$\int_0^t \int_0^t \delta(X_s - X_\sigma) ds d\sigma = t^2 \int_{\mathbb{R}} [l_{t,u,\omega}]^2 du$$

where $l_{t,u,\omega}$ is the density of the occupation measure $L_t(\omega)$ with respect to the Lebesgue measure on $\mathbb{R}$:

$$L_t(\omega, A) = \frac{1}{t} \int_0^t 1_A(\omega_s) ds \quad \text{for } A \in \sigma(\mathbb{R}).$$

Let $M_{\mathbb{R}}$ be the set of all probability measures on $\mathbb{R}$ equipped with the topology of weak convergence. We define a functional $\Phi: M_{\mathbb{R}} \rightarrow [0, \infty]$ by

$$\Phi(\mu) = \begin{cases} \int_{\mathbb{R}} \left[\frac{d\mu}{d\lambda} \right]^2 d\lambda & \text{for } \mu \ll \lambda \text{ and } \frac{d\mu}{d\lambda} \in L^2(\mathbb{R}) \\ \infty & \text{otherwise.} \end{cases} \quad (5.2)$$

LEMMA 5.3 Φ is a lower semi-continuous, strictly convex functional on $M_{\mathbb{R}}$.

Proof We have to show that in the case of $\Phi(\mu) < \infty$ for every weakly convergent sequence $\mu_n \rightarrow \mu$ it holds

$$\liminf_{n \rightarrow \infty} \Phi(\mu_n) \geq \Phi(\mu).$$

Look at $C_c(\mathbb{R})$, the set of continuous functions with compact support. $C_c(\mathbb{R})$ is a dense subspace of $L^2(\mathbb{R})$ and for every $f \in L^2(\mathbb{R})$ we have $\|f\|_2 = \sup_{\phi \in C_c(\mathbb{R}); \|\phi\|_2 = 1} \langle \phi, f \rangle$. The functional $f \rightarrow \langle \phi, f \rangle$ is a continuous functional on $L^2(\mathbb{R})$ for all $\phi \in C_c(\mathbb{R})$. Therefore we get that $f \rightarrow \sup_{\phi \in C_c(\mathbb{R}); \|\phi\|_2 = 1} \langle \phi, f \rangle$ is a lower semi-continuous functional.

Now we see that in the case where $\Phi(\mu) < \infty$ and $\{\mu_{n_j}\}_{j \in \mathbb{N}}$ a subsequence of $\{\mu_n\}_{n \in \mathbb{N}}$ with $d\mu_{n_j}/d\lambda = f_{n_j} \in L^2(\mathbb{R})$ for all $j \in \mathbb{N}$ the weak convergence of $\{\mu_{n_j}\}_{j \in \mathbb{N}}$ gives us $\lim_{j \rightarrow \infty} \|f_{n_j}\|_2^2 \geq \|f\|_2^2$ where $d\mu/d\lambda = f \in L^2(\mathbb{R})$.

We have still to show that in the case of $\Phi(\mu) = \infty$ it holds that for every subsequence $\{\mu_{n_j}\}_{j \in \mathbb{N}}$ of $\{\mu_n\}_{n \in \mathbb{N}}$ with $d\mu_{n_j}/d\lambda = f_{n_j} \in L^2(\mathbb{R})$ it holds $\lim_{j \rightarrow \infty} \Phi(\mu_{n_j}) = \infty$. But this is a consequence of the Banach-Alaoglu theorem. Suppose there exists a subsequence $\{\mu_{n_j}\}_{j \in \mathbb{N}}$ of $\{\mu_n\}_{n \in \mathbb{N}}$ with $d\mu_{n_j}/d\lambda = f_{n_j} \in L^2(\mathbb{R})$ such that $\lim_{j \rightarrow \infty} \Phi(\mu_{n_j}) < \infty$. Then this sequence must have a limit point f^* in $L^2(\mathbb{R})$ with $\|f^*\|_2 < \infty$. But this leads to a contradiction because of the equality: $\int \phi f^* d\lambda = \lim_{j \rightarrow \infty} \int \phi d\mu_{n_j} = \int \phi d\mu$ for all $\phi \in C_c(\mathbb{R})$. Using the remark about the L^2 -norm we find $\Phi(\mu) = \|f^*\|_2^2 < \infty$. This shows Lemma 5.3. ■

Using Lemma 5.3 and Lemma 2.1.7 of Deuschel and Stroock [2] we get

THEOREM 5.4 (Lower Estimate)

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \left\{ \frac{2}{t} \int_0^t \int_0^t \delta(X_s - X_\sigma) ds d\sigma \right\} \geq \sup_{\substack{\phi \in L^2_{\mathbb{R}} \\ \|\phi\|_2 = 1}} [2|\phi|_4^4 - \frac{1}{2}|\phi'|_2^2].$$

In order to prove the upper estimate we have to go through several steps. We first need to approximate Φ by smooth functionals Φ_ε and second calculate an upper estimate for the Φ_ε on the torus $\mathbb{R}/m\mathbb{Z}$, $m > 0$. This will allow us to get the estimate for Φ on $\mathbb{R}$.

Let $k(x) = \max\{0, 1 - |x|\}$. For $\varepsilon > 0$ we define the function $k_\varepsilon: \mathbb{R} \rightarrow \mathbb{R}$ by $k_\varepsilon(x) = \varepsilon^{-1}k(\varepsilon^{-1}x)$. The function k is the density of a probability measure and the Fourier transforms of k and k_ε are given by

$$\widehat{k}(\lambda) = \frac{2(\cos \lambda - 1)}{\lambda^2} \quad \text{and} \quad \widehat{k_\varepsilon}(\lambda) = \widehat{k}(\varepsilon \lambda).$$

We define functionals $\Phi_\varepsilon: M_{\mathbb{R}} \rightarrow [0, \infty[$ by

$$\Phi_\varepsilon(\mu) = \|\mu_\varepsilon\|_2^2 \quad \text{and} \quad \mu_\varepsilon(x) = k_\varepsilon * \mu(x) = \int_{\mathbb{R}} k_\varepsilon(x - y) \mu(dy).$$

LEMMA 5.5 Φ_ε is a continuous and bounded functional on $M_{\mathbb{R}}$.

Proof The boundedness follows by direct calculation. We show only the continuity. The weak convergence $\mu_n \rightarrow \mu$ implies tightness of the family $\{\mu; \mu_n, n \in \mathbb{N}\}$ and equicontinuity of the family of functions $\{\mu_\varepsilon(x); \mu_n^\varepsilon(x), n \in \mathbb{N}\}$. Furthermore we have:

$$|\Phi_\varepsilon(\mu_n) - \Phi_\varepsilon(\mu)| = \left| \int_{\mathbb{R}} [\mu_n^\varepsilon(x) - \mu^\varepsilon(x)] [\mu_n^\varepsilon(x) + \mu^\varepsilon(x)] dx \right| \leq 2 \sup_{x \in \mathbb{R}} |\mu_n^\varepsilon(x) - \mu^\varepsilon(x)|$$

$$|\mu_n^\varepsilon(x) - \mu^\varepsilon(x)| = \left| \int_{\mathbb{R}} k_\varepsilon(x-y)(\mu_n(dy) - \mu(dy)) \right| \leq \varepsilon^{-1} |\mu_n(I_\varepsilon^x) - \mu(I_\varepsilon^x)|$$

with $I_\varepsilon^x = [x - \varepsilon; x + \varepsilon]$. We conclude that $|\mu_n(I_\varepsilon^x) - \mu(I_\varepsilon^x)| \rightarrow 0$ uniformly on $\mathbb{R}$ where we used the fact that $\nu_\varepsilon(x)$ is uniformly continuous for every $\nu \in M_{\mathbb{R}}$. ■

LEMMA 5.6 $\Phi_\varepsilon \leq \Phi$ for all $\varepsilon > 0$ and $\Phi_\varepsilon \rightarrow \Phi$ for $\varepsilon \rightarrow 0$.

Proof Case 1: $\Phi(\mu) < \infty$. The statement of the lemma follows directly from the properties of the Fourier transform of k_ε and the fact that the Fourier transform is an L^2 -isometry.

Case 2: $\Phi(\mu) = \infty$. We have to show that $\Phi_\varepsilon(\mu) \rightarrow \infty$ for $\varepsilon \rightarrow 0$. We first note that $\Phi_\varepsilon(\mu) = \Phi(\mu_\varepsilon)$. If $\Phi_\varepsilon(\mu) \leq c < \infty$ for all $\varepsilon > 0$ then we can apply the Banach-Alaoglu theorem and get a weakly convergent subsequence denoted by $\{\mu_{\varepsilon_k}\}$ with μ as a limit point. This follows from the weak convergence of $\mu_{\varepsilon_k} \rightarrow \mu$ and contradicts the fact that Φ is lower semi-continuous. This proves Lemma 5.6. ■

For $m > 0$ let T_m be the "circle" of size m , i.e. the set of equivalence classes defined by the equivalence relation $x \sim y$ if $x, y \in \mathbb{R}$ and $x - y$ is an integral multiple of m . Let g_m be the canonical map $\mathbb{R} \rightarrow T_m$, i.e. any element of $\mathbb{R}$ is mapped by g_m onto the equivalence class to which it belongs. It is possible to extend g_m naturally to the path space. Now we can define Brownian motion on T_m by the measure $P_m = P \circ g_m^{-1}$. We define also in an obvious manner the occupation measure on $T_m: L_t^m(\omega^*, \cdot)$. Let M_m be the set of probability measures on T_m . For $\varepsilon > 0$ we define functionals $\Phi_{\varepsilon, m}: M_m \rightarrow [0, \infty]$ by

$$\Phi_{\varepsilon, m}(\mu) = \int_{T_m} [\mu_\varepsilon(x)]^2 dx \quad \text{with} \quad \mu_\varepsilon(x) = \int_{T_m} k_\varepsilon * (x - y) \mu(dy)$$

with $k_\varepsilon * (x) = \sum_{n \in \mathbb{Z}} k_\varepsilon(x + nm)$. $\Phi_{\varepsilon, m}$ is a continuous bounded functional on M_m .

LEMMA 5.7 For $m > 0$ and small ε it holds:

$$E^P \exp \{2t \Phi_\varepsilon(L_t(\omega; \cdot))\} \leq E^{P^m} \exp \{2t \Phi_{\varepsilon, m}(L_t^m(\omega^*, \cdot))\}.$$

Proof

$$\begin{aligned} E^P \exp \{2t \Phi_\varepsilon(L_t(\omega; \cdot))\} &\leq E^P \exp \left\{ 2t \int_{\mathbb{R}} [L_t^\varepsilon(\omega, x)]^2 dx \right\} \\ &\leq E^P \exp \left\{ 2t \sum_{k \in \mathbb{Z}} \int_{km}^{(k+1)m} [L_t^\varepsilon(\omega, x)]^2 dx \right\} \\ &\leq E^P \exp \left\{ 2t \int_0^m \left[\sum_{k \in \mathbb{Z}} \int_{\mathbb{R}} k_\varepsilon(x - (y + km)) L_t(\omega, dy) \right]^2 dx \right\} \end{aligned}$$

$$\leq E^{P_m} \exp \left\{ 2t \int_0^m \left[\sum_{k \in \mathbb{Z}} L_t^\varepsilon(g_m^{-1}(\omega^*), x + km) \right]^2 dx \right\}$$

$$\leq E^{P_m} \exp \{ 2t \Phi_{\varepsilon, m}[L_t^m(\omega^*, \cdot)] \}. \quad \blacksquare$$

The Large Deviation theory as presented in Donsker and Varadhan [3] yields

$$\overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^{P_m} \exp \{ 2t \Phi_{\varepsilon, m}[L_t^m(\omega^*, \cdot)] \} \leq \sup_{\mu \in M_m} \{ 2\Phi_{\varepsilon, m}(\mu) - I_m(\mu) \}$$

with

$$I_m(\mu) = \frac{1}{8} \int_{T_m} \frac{[f'(x)]^2}{f(x)} dx \quad \text{for} \quad f = \frac{d\mu}{d\lambda_m} \quad \text{otherwise} \quad I_m(\mu) = \infty.$$

LEMMA 5.8

$$\overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \{ 2t \Phi_\varepsilon[L_t(\omega, \cdot)] \} \leq \sup_{\mu \in M_{\mathbb{R}}} \{ 2\Phi_\varepsilon(\mu) - I(\mu) \}$$

Proof Lemma 5.7 combined with the above citation of the large deviation result from Donsker and Varadhan [3] yields for all $m > 1$:

$$\overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \{ 2t \Phi_\varepsilon[L_t(\omega, \cdot)] \} \leq \sup_{\mu \in M_m} \{ 2\Phi_{\varepsilon, m}(\mu) - I_m(\mu) \}.$$

The lemma will be proved if we can show that

$$\sup_{\mu \in M_{\mathbb{R}}} \{ 2\Phi_\varepsilon(\mu) - I(\mu) \} \geq \inf_{m > 1} \sup_{\mu \in M_m} \{ 2\Phi_{\varepsilon, m}(\mu) - I_m(\mu) \}$$

This follows by using the ideas of Lemma 2.2 of Section 2. As done in Step 1 of the proof of this lemma, we construct to a given arbitrarily large $m > 1$ and a given probability measure $\mu \in M_m$ a measure $\mu^* \in M_{\mathbb{R}}$ such that $I_m(\mu)$ is nearly the same as $I(\mu^*)$. For this pair μ and μ^* we also have to show that $|\Phi_{\varepsilon, m}(\mu) - \Phi_\varepsilon(\mu^*)|$ is small too. But this is easily seen from the construction of μ^* . $\blacksquare$

LEMMA 5.9 For every $q > 1$ it holds:

$$\overline{\lim}_{\varepsilon \rightarrow 0} \overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \left\{ 4tq \sup_{|x-y| < \varepsilon} |l_{t,x} - l_{t,y}| \right\} = 0.$$

Proof Let $\gamma > 0$ (small enough) be given, then we have the estimate

$$\begin{aligned}
& E^P \exp \left\{ 4tq \sup_{|x-y| < \varepsilon} |l_{t,x} - l_{t,y}| \right\} \\
& \leq \exp \{ 4tq\gamma \} + \sum_{k=0}^{\infty} \exp \{ 4tq(\gamma + k + 1) \} P \left\{ \sup_{|x-y| < \varepsilon} |l_{t,x} - l_{t,y}| > \gamma + k \right\}.
\end{aligned}$$

The lemma will follow, if we can show that for a small $\varepsilon > 0$

$$\overline{\lim}_{t \rightarrow \infty} \sum_{k=0}^{\infty} \exp \{ 4tq(\gamma + k + 1) \} P \left\{ \sup_{|x-y| < \varepsilon} |l_{t,x} - l_{t,y}| > \gamma + k \right\}$$

will be small. Therefore we look at the probabilities $P\{\sup_{|x-y| < \varepsilon} |l_{t,x} - l_{t,y}| > a\}$ for $a > 0$. The scaling property of Brownian motion yields

$$P \left\{ \sup_{|x-y| < \varepsilon} |l_{t,x} - l_{t,y}| > a \right\} = P \left\{ \sup_{|x-y| < \varepsilon/\sqrt{t}} |l_{1,x} - l_{1,y}| > a\sqrt{t} \right\}.$$

Applying Lemma 3.4 of Donsker and Varadhan [7] which states a result by Garsia, Rodemich and Rumsey as well as Lemma 3.3 of Donsker and Varadhan [7] we are able to use the proof given to show Lemma 3.5 and (3.4.2) of Section 3. Going through the same steps except for some obvious modifications we get:

For $\gamma > 0$ and ε small enough we have

$$\overline{\lim}_{t \rightarrow \infty} \sum_{k=0}^{\infty} \exp \{ 4tq(\gamma + k + 1) \} P \left\{ \sup_{|x-y| < \varepsilon} |l_{t,x} - l_{t,y}| > \gamma + k \right\} \leq \gamma.$$

This proves Lemma 5.9. $\blacksquare$

THEOREM 5.10 (Upper Estimate)

$$\overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^P \exp \{ 2t\Phi(L_t(\omega, \cdot)) \} \leq \sup_{\substack{\phi \in L^2(\mathbb{R}) \\ \|\phi\|_2 = 1}} [2\Phi(\phi^2) - \frac{1}{2}\|\phi'\|_2^2]$$

Proof Using Hölder's inequality gives us

$$\begin{aligned}
E^P \exp \{ 2t\Phi(L_t(\omega, \cdot)) \} &= E^P \exp \{ 2t[\Phi(L_t(\omega, \cdot)) - \Phi_\varepsilon(L_t(\omega, \cdot)) + \Phi_\varepsilon(L_t(\omega, \cdot))] \} \\
&\leq \sqrt[q]{E^P \exp \{ 2tq[\Phi(L_t(\omega, \cdot)) - \Phi_\varepsilon(L_t(\omega, \cdot))] \}} \sqrt[p]{E^P \exp \{ 2tp\Phi_\varepsilon(L_t(\omega, \cdot)) \}}.
\end{aligned}$$

For every $p > 1$ and conjugated q this implies:

$$\begin{aligned} \overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^p \exp \{2t\Phi(L_t(\omega, \cdot))\} &\leq \frac{1}{p} \overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^p \exp \{2tp\Phi_\varepsilon(L_t(\omega, \cdot))\} \\ &+ \frac{1}{q} \overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^p \exp \{2tq[\Phi(L_t(\omega, \cdot)) - \Phi_\varepsilon(L_t(\omega, \cdot))]\}. \end{aligned}$$

We show

$$\overline{\lim}_{\varepsilon \rightarrow 0} \overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^p \exp \{2tq[\Phi(L_t(\omega, \cdot)) - \Phi_\varepsilon(L_t(\omega, \cdot))]\} = 0$$

for all $q > 1$. But this is a direct consequence of Lemma 5.9 because of the relations:

$$|\Phi - \Phi_\varepsilon|(L_t(\omega, \cdot)) = \int_{\mathbb{R}} [l_{t,x,\omega}]^2 - [k_\varepsilon * l_{t,\cdot,\omega}]^2 dx \leq 2 \sup_{|x-y| < \varepsilon} |l_{t,x} - l_{t,y}|.$$

Now we have

$$\overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^p \exp \{2t\Phi(L_t(\omega, \cdot))\} \leq \overline{\lim}_{\varepsilon \rightarrow 0} \frac{1}{p} \overline{\lim}_{t \rightarrow \infty} \frac{1}{t} \log E^p \exp \{2tp\Phi_\varepsilon(L_t(\omega, \cdot))\}$$

for all $p > 1$. Using the result of Lemma 5.6 which says that $\Phi_\varepsilon \leq \Phi$ and the function $S(p)$ introduced in Step 3 of the proof of Theorem 4.1 and going through the argumentation there, proves Theorem 5.10. ■

6. SOLUTION OF THE VARIATIONAL PROBLEM

Let $W^{2,1}(\mathbb{R})$ be the space of all measurable functions $\phi: \mathbb{R} \rightarrow \mathbb{R}$ where ϕ and ϕ' are elements of $L^2(\mathbb{R})$. Look at the functional $E: W^{2,1}(\mathbb{R}) \rightarrow \mathbb{R}$ defined as $E(\phi) = 2|\phi|_4^4 - \frac{1}{2}|\phi'|_2^2$. In this section we show:

THEOREM 6.1 *Let $M = \sup\{E(\phi) | \phi \in W^{2,1}(\mathbb{R}) \text{ and } |\phi|_2 \leq 1\}$, then it holds:*

$$M \text{ is finite and positive.} \quad (6.1.1)$$

$$\begin{aligned} &\text{There exists a smooth real function } \varphi \text{ with } |\varphi|_2 = 1, E(\varphi) = M \text{ and} \\ &\text{every such maximizing function fulfils the nonlinear equation} \\ &[\tfrac{1}{2}\Delta + 4|\varphi|^2]\varphi = e\varphi \text{ for an } e > 0. \end{aligned} \quad (6.1.2)$$

$$\varphi \text{ is uniquely determined up to translation.} \quad (6.1.3)$$

$$\varphi(x) = \frac{1}{\sqrt{2}} \operatorname{sech}(x) = \frac{\sqrt{2}}{\exp(-2x) + \exp(2x)}. \quad (6.1.4)$$

The proof of Theorem 6.1 will be a direct consequence of the following results.

LEMMA 6.2

For all $\phi \in W^{2,1}(\mathbb{R})$ with $|\phi|_2 \leq 1$ it holds that $E(\phi) \leq c < \infty$ and c independent of ϕ . (6.2.1)

$$M > 0, \quad (6.2.2)$$

For $\phi \in W^{2,1}(\mathbb{R})$ with $E(\phi) > M - 1$ it holds that $|\phi'|_2 \leq c < \infty$ and c independent of ϕ . (6.2.3)

For $\phi \in W^{2,1}(\mathbb{R})$ with $E(\phi) = M$ and $|\phi|_2 \leq 1$ it holds that $|\phi|_2 = 1$. (6.2.4)

Proof of Lemma 6.2 First we show (6.2.1):

$$\begin{aligned} E(\phi) &= 2|\phi|_4^4 - \frac{1}{2}|\phi'|_2^2 \\ &\leq 2 \left[\operatorname{ess\,sup}_{x \in \mathbb{R}} |\phi| \right]^2 |\phi|_2^2 - \frac{1}{2}|\phi'|_2^2 \leq 2 \left[\operatorname{ess\,sup}_{x \in \mathbb{R}} |\phi| \right]^2 - \frac{1}{2}|\phi'|_2^2. \end{aligned}$$

For $\phi \in C^1(\mathbb{R})$ we can apply the fundamental theorem of calculus and get for $\varepsilon > 0$ and $y \in [x, x + \varepsilon]$:

$$\begin{aligned} \varepsilon |\phi(x)| &\leq \int_{\mathbb{R}} 1_{[x, x+\varepsilon]}(y) |\phi(y)| dy + \int_{\mathbb{R}} \left\{ 1_{[x, x+\varepsilon]}(y) \int_x^y |\phi'(u)| du \right\} dy \\ &\leq \sqrt{\varepsilon} |\phi|_2 + \varepsilon \sqrt{\varepsilon} |\phi'|_2. \end{aligned}$$

For $\phi \in C^1(\mathbb{R})$ it follows

$$\sup_{x \in \mathbb{R}} |\phi(x)| \leq \varepsilon^{-0.5} |\phi|_2 + \sqrt{\varepsilon} |\phi'|_2. \quad (6.2.5)$$

Now we set $\varepsilon = 1/9$ and get

$$\sup_{x \in \mathbb{R}} |\phi(x)| \leq 3|\phi|_2 + \frac{1}{3}|\phi'|_2.$$

It is possible to approximate every $\phi \in W^{2,1}(\mathbb{R})$ by a sequence of smooth functions $\{\phi_n\}_{n \in \mathbb{N}}$ (Adams [1, Theorem 3.16]). $\{\phi_n\}_{n \in \mathbb{N}}$ is a $W^{2,1}(\mathbb{R})$ -Cauchy sequence and by (6.2.5) it follows, that $\{\phi_n\}_{n \in \mathbb{N}}$ must converge towards a continuous function. Therefore ϕ must be essentially a $C_b(\mathbb{R})$ -function. This shows that

$$\operatorname{ess\,sup}_{x \in \mathbb{R}} |\phi(x)| \leq 3|\phi|_2 + \frac{1}{3}|\phi'|_2. \quad (6.2.6)$$

From here we get the estimate

$$E(\phi) \leq 18 + 8|\phi'|_2 - \frac{1}{18}|\phi'|_2^2 < c < \infty.$$

This proves (6.2.1). The proof of (6.2.2) follows directly from a special choice of ϕ : With $\phi(x) = (2\pi)^{-0.25} \exp\{-\frac{1}{4}x^2\}$ we have $E(\phi) = (1/\sqrt{\pi}) - \frac{1}{8} > 0$. The proof of (6.2.3) follows by a similar argumentation as used to prove (6.2.1): Let $\phi \in W^{2,1}(\mathbb{R})$ with $|\phi|_2 \leq 1$ and $E(\phi) > M - 1$. This gives us:

$$2 \left| \operatorname{ess\,sup}_{x \in \mathbb{R}} |\phi(x)| \right|^2 + 1 - M \geq 2|\phi|_4^4 + 1 - M \geq \frac{1}{2}|\phi'|_2$$

using (6.2.6) we get $19 - M + 4|\phi'|_2 - \frac{5}{18}|\phi'|_2^2 \geq 0$. Looking at the graph of the parabel $19 - M + 4x - 5/18x^2$ and the fact that $|\phi'|_2$ has to be non-negative we get (6.2.3). The proof of (6.2.4) follows from an easy calculation: Let $\phi_0 \in W^{2,1}(\mathbb{R})$ with $|\phi_0|_2 = \gamma < 1$ and $E(\phi_0) = M$. Define $\psi = \gamma^{-1}\phi_0$ then we have

$$E(\psi) = \gamma^{-2} [2\gamma^{-2}|\phi_0|_4^4 - \frac{1}{2}|\phi_0'|_2^2] \geq \gamma^{-2} E(\phi_0) \geq \gamma^{-2} M > M$$

which is a contradiction and therefore proves (6.2.4). Now Lemma 6.2 is proved. ■

Remark 6.2.7 Using the notation

$$S_m(\phi) = a \int_{\mathbb{R}} \int_{\mathbb{R}} \phi(x)^2 f^m(x-y) \phi(y)^2 dx dy - \frac{1}{2} |\phi'(y)|_2^2 \quad \text{and} \quad S_m = \sup_{\substack{\phi \in L^2(\mathbb{R}) \\ |\phi|_2 = 1}} [S_m(\phi)]$$

we can go through the same calculation as done in the proof of the lemma and show that for $\phi \in W^{2,1}(\mathbb{R})$ with $|\phi|_2 \leq 1$ and all $m \in \mathbb{N}$ we have $S_m(\phi) \leq c < \infty$ where c is independent of ϕ .

The proof is based on the relation:

$$\begin{aligned} S_m(\phi) &\leq a \left[\operatorname{ess\,sup}_{x \in \mathbb{R}} |\phi| \right]^2 \int_{\mathbb{R}} f^m * \phi^2(x) dx - \frac{1}{2} |\phi|_2^2 \\ &\leq a \left[\operatorname{ess\,sup}_{x \in \mathbb{R}} |\phi| \right]^2 - \frac{1}{2} |\phi|_2^2, \end{aligned}$$

where we used the convolution estimate $|f * g|_1 \leq |f|_1 \cdot |g|_1$. Now we are able to follow the same arguments given in the proof of the lemma above.

Remarks About "Symmetric" Decreasing Rearrangements 6.3 The following notation is taken from Lieb [12, Section III]. Let S be the set

$$S = \{f: \mathbb{R} \rightarrow [0, \infty], f(x) \leq f(y) \text{ for } |x| \geq |y| \text{ and } f(x) = f(-x)\}.$$

S is a set of Lebesgue measurable functions.

If 1_A is the characteristic function of a measurable set A in $\mathbb{R}$ then we define 1_A^* as $1_A^* = 1$ if $|x| \leq 1/2\lambda(A)$ and $1_A^* = 0$ otherwise. $1_A^* \in S$ and $|1_A^*|_1 = |1_A|_1$. Given $f: \mathbb{R} \rightarrow [0, \infty]$ let 1_a^f be defined by $1_a^f(x) = 1$ for $f(x) \geq a$, $1_a^f(x) = 0$ otherwise.

Then it holds

$$f(x) = \int_0^\infty 1_a^f(x) ds \quad \text{and we define } f^* \text{ by } f^*(x) = \int_0^\infty [1_a^f]^*(x) da.$$

Clearly: $f^* \in S$.

$$\lambda\{x: f^*(x) \geq a\} = \lambda\{x: f(x) \geq a\} \quad \lambda \text{ the Lebesgue measure on } \mathbb{R}.$$

$$|f^*|_p = |f|_p \text{ for } p \geq 1.$$

$$\text{For } \alpha > 0 \quad (f^*)^\alpha = (f^\alpha)^* \text{ pointwise.}$$

LEMMA 6.4 For $\phi \in W^{2,1}(\mathbb{R})$ it follows that $\phi^* \in W^{2,1}(\mathbb{R})$ with $|\phi'|_2 > |\phi^*|_2$ unless there exists a $y \in \mathbb{R}$ such that $\phi(x-y) = \phi^*(x)$ a.e.

Proof of Lemma 6.4 We follow the proof of Lemma 5 in the Appendix of Lieb [12]. There Lieb studies the expression:

$$I_t(\phi) = t^{-1} \left\{ \int_{\mathbb{R}} |\phi(x)|^2 dx - \int_{\mathbb{R}} \phi(x) G_t(x-y) \phi(y) dx dy \right\}.$$

The function G_t is given by $G_t(x) = (4\pi t)^{-1} \exp\{-x^2/4t\}$. Lemma 6.4 will be shown, if it holds that $I_t(\phi) > I_t(\phi^*)$. Applying the "strong version" of the Riesz theorem as stated by Lieb [12, Lemma 3 in the Appendix] we get:

$$\int_{\mathbb{R}} \phi(x) G_t(x-y) \phi(y) dx dy < \int_{\mathbb{R}} \phi^*(x) G_t(x-y) \phi^*(y) dx dy$$

unless the following holds: There exists a $y \in \mathbb{R}$ such that $\phi(x-y) = \phi^*(x)$ a.e. This gives the relation $I_t(\phi) > I_t(\phi^*)$ and the following the proof of Lemma 5 in Lieb [12] we get $|\phi'|_2 > |\phi^*|_2$. This proves Lemma 6.4. ■

Remark The strict version of the inequality is needed to show that a maximizing function is necessarily symmetric.

LEMMA 6.5 For $\phi \in W^{2,1}(\mathbb{R})$ it holds $E(\phi) < E(\phi^*)$.

LEMMA 6.6 There exists a sequence of symmetric decreasing $W^{2,1}(\mathbb{R})$ -functions $\{\phi_j\}_{j \in \mathbb{N}}$ with $|\phi_j|_2 = 1$ and $E(\phi_j) \uparrow M$.

Both lemmata follow from Lemma 6.4 and Lemma 6.2 and Remarks 6.3.

THEOREM 6.7 (Existence of a Maximizing Function) There exists a $\phi \in S$ with $|\phi|_2 = 1$ and $E(\phi) = M$.

Proof of 6.7 Let $\{\phi_j\}_{j \in \mathbb{N}}$ be a maximizing sequence of S -functions. Every subsequence of $\{\phi_j\}_{j \in \mathbb{N}}$ will also be a maximizing sequence of S -functions. $W^{2,1}(\mathbb{R})$ is a Hilbert space with norm $|\phi|_W = |\phi|_2 + |\phi'|_2$. From (6.2.3) we know that $|\phi'|_2$ is

bounded. This shows that $\{\phi_j\}_{j \in \mathbb{N}}$ lies in a closed ball of the Hilbert space $W^{2,1}(\mathbb{R})$. By the Banach-Alaoglu theorem there exists a $W^{2,1}(\mathbb{R})$ -weakly convergent subsequence also denoted by $\{\phi_j\}_{j \in \mathbb{N}}$. Let ϕ be a weak limit point of the sequence. We will show that ϕ is a candidate for the maximizing function, i.e. $E(\phi) = M$. If ϕ is the weak limit point then $\liminf |\phi'_j|_2^2 \geq |\phi'|_2^2$ and $|\phi|_2^2 \leq 1$. Both inequalities are a direct consequence of the weak convergence and the continuity of the inner product in a Hilbert space. If we are able to show that ϕ is a maximizing function then it follows by (6.2.4) that $|\phi|_2 = 1$. So far we know:

$$E(\phi_j) = 2|\phi_j|_4^4 - \frac{1}{2}|\phi'_j|_2^2 \leq 2|\phi_j|_4^4 - \frac{1}{2}|\phi'|_2^2$$

and ϕ will be a maximizing function as soon as we have:

$$\lim_{j \rightarrow \infty} |\phi_j|_4^2 = |\phi|_4^2.$$

This will be done using the dominated convergence theorem in the following form: (i) $\psi_j(x) \rightarrow \psi(x)$ pointwise Lebesgue a.e.; (ii) $\psi_j \in L^4(\mathbb{R})$; (iii) $\psi_j^4 \leq f(x)$ with $f \in L^1(\mathbb{R})$ then it holds $\lim_{j \rightarrow \infty} |\psi_j|_4^4 = |\psi|_4^4$. First we note that ϕ_j converges weakly to ϕ and not pointwise λ -a.e. Therefore we have to find a pointwise convergent subsequence of $\{\phi_j\}$ which we will denote by $\{\phi_j^*\}$. Now consider $\{\phi_j^2\}_{j \in \mathbb{N}}$ which is a sequence of densities of probability measures. Using (6.2.6) and (6.2.3) we get that $\phi_j^2(0) \leq c' < \infty$ for all $j \in \mathbb{N}$. Because $\phi_j \in S$, the graph of the function ϕ_j shows a special shape which allows us to apply Helly's theorem as stated in Breiman, Probability, p. 160, to functions of the form:

$$\psi:]-\infty, 0] \rightarrow [0, c'] \text{ with } x \leq y \Rightarrow \psi(x) \leq \psi(y) \text{ and } x \uparrow y \Rightarrow \psi(x) \uparrow \psi(y).$$

Now we can find a further subsequence $\{\phi_j^*\}_{j \in \mathbb{N}}$ of $\{\phi_j\}_{j \in \mathbb{N}}$ such that $(\phi_j^*)^2 \rightarrow (\phi^*)^2$ pointwise on the set points of continuity of ϕ^* . $\phi^* \in S$ gives the pointwise convergence λ -a.e. looked for. Now we construct the dominating function for the $\{\phi_j^*\}$ sequence. Using the fact that ϕ_j^* is a symmetric function and an element of S and applying the Sobolev Imbedding Corollary 5.6 of Adams [1] we get:

$$\phi_j^{*8}(x) \cdot x \leq \int_0^x \phi_j^{*8}(u) \cdot 1 \, du \leq \int_{\mathbb{R}} \phi_j^{*8}(u) \, du \leq |\phi_j^*|_8^8 < \infty$$

$$\phi_j^{*2}(x) \cdot x \leq |\phi_j^*|_2^2 = 1.$$

This gives the dominating function:

$$\phi_j^{*4}(x) \leq f(x) \text{ with } f \text{ defined by } f(x) = \begin{cases} A|x|^{-0,5} & \text{for } |x| \leq 1 \\ A|x|^{-2} & \text{for } |x| \geq 1 \end{cases} \text{ and } f \in L^1(\mathbb{R}).$$

We apply the Lebesgue theorem to $\{\phi_j^*\}$ and it yields: $\lim_{j \rightarrow \infty} |\phi_j^*|_4^4 = |\phi^*|_4^4$.

Everything is shown if we have $|\phi^*|_4^4 = |\phi|_4^4$. Therefore we show that $\phi = \phi^*$ λ -a.e. This is followed again by the dominated convergence theorem and the fact that $\sqrt[4]{f} \in L_{loc}^1(\mathbb{R})$. Using weak convergence gives us $\int g(\phi_j^* - \phi) d\lambda \rightarrow 0$ for every smooth real function with compact support. Using the Lebesgue theorem yields $\int g(\phi_j^* - \phi^*) d\lambda \rightarrow 0$. This gives us $\phi = \phi^*$ λ -a.e. and proves Theorem 6.7. ■

THEOREM 6.8

$\phi \in W^{2,1}(\mathbb{R})$, $|\phi|_2 = 1$ and $E(\phi) = M$. Then ϕ satisfies the weak equation: $\{\frac{1}{2}\Delta + 4|\phi(x)|^2\}\phi(x) = \lambda\phi(x)$ for some $\lambda > 0$. (6.8.1)

For $\phi \in W^{2,1}(\mathbb{R})$ and ϕ satisfying the equation in (6.8.1) for a $\lambda > 0$ it holds that ϕ is a smooth function and $\phi(x) \rightarrow 0$ for $|x| \rightarrow \infty$. Therefore ϕ is also a strong solution of the equation in (6.8.1). (6.8.2)

Proof of Theorem 6.8 (6.8.1) follows by taking the Fréchet-derivative $\lim_{h \rightarrow 0} (1/h)[E(\phi + hg) - E(\phi)] = 0$ under the restriction $\langle \phi, g \rangle_{L_2} = 0$. This gives $4\phi^2 \cdot \phi + \frac{1}{2}\phi'' = \lambda\phi$. Multiplying both sides by ϕ and integrating yields $E(\phi) = \lambda - 2|\phi|_4^4$ which shows that $\lambda > 0$. The proof of (6.8.2) follows by using part (iii) of the proof of Theorem 8 in Lieb [12]. But we work with the kernel $K_\lambda(x) = 1/(2\sqrt{\lambda}) \exp\{-\sqrt{\lambda}|x|\}$. This finishes the proof of Theorem 6.8. ■

LEMMA 6.9 The function $\phi(x) = 1/\sqrt{2} \operatorname{sech}(x) = \sqrt{2}/(\exp\{-2x\} + \exp\{2x\})$ is a modulo translation of the uniquely determined maximizing function for the functional E with $\phi \in W^{2,1}(\mathbb{R})$ and $|\phi|_2 = 1$.

Proof of Lemma 6.9 We have to find a function ϕ with the following properties: (1) ϕ satisfies the ODE $4\phi^3 + \frac{1}{2}\phi'' = \lambda\phi$ for a $\lambda > 0$. (2) ϕ is an even function. (3) ϕ is a smooth function and $\phi'(0) = 0$ (because of (2)). (4) ϕ is non-negative and monotonically decreasing on $[0, \infty[$ with $\phi(x) \rightarrow 0$ for $x \rightarrow \infty$. (5) ϕ^2 is a density of a probability measure. The ODE given in (1) can be solved explicitly and uniquely under restrictions (2)–(5). This proves Lemma 6.9. ■

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